Do ESG investors care about carbon emissions? Evidence from securitized auto loans*

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Abstract

Over the last decade, assets under management using environmental, social, and governance (ESG) criteria have increased tenfold. This paper shows that financial markets *do not* price differences in CO_2 emissions of auto loans despite the rise of ESG investing. I argue that securitized auto loans present a clean empirical setting to measure ESG preferences in price and quantity data. I find that securities with high-emission pools have, on average, 5 basis points (11%) lower cost of capital. Mutual funds marketed as ESG (i) hold positions across the full distribution of CO_2 emissions and (ii) invest more in higher-emission deals compared to non-ESG funds. Neither (i) nor (ii) are consistent with common ESG strategies.

JEL classification: G12, G18, G20, G41, Q56

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1 Introduction

Over the last decade, assets under management using environmental, social, and governance (ESG) criteria have increased tenfold. Money managers report that the leading ESG criteria are climate change and carbon emissions.[1] This large inflow of capital suggests an appetite from investors for greener assets. High demand for green assets could lower the cost of capital for green firms (e.g., as in Heinkel et al., 2001). That a lower cost of capital for greener firms is due to ESG preferences, however, requires that firms and investors do not engage in "greenwashing," and that assets that are marketed as green are not simply assets offering higher expected cash flows. Estimating whether investors truly have a preference for greenness therefore faces a major challenge: in almost all settings, it is difficult to hold cash flow risk constant while varying ESG-relevant characteristics. This is the case for two reasons. First, it is impossible to keep cash flows in all states of the world constant across securities. Second, many ESG criteria influence cash flows across states of the world (e.g., via carbon taxation). The correlation between cash flows and ESG criteria can introduce unobserved heterogeneity across securities, limiting causal inference. Questions about data reliability of ESG rating providers (Berg et al., 2020), which strategy ESG investors follow, and whether a given activity is ESG or greenwashing further complicate causal inference.

This paper addresses these challenges and sheds new light on whether green assets have a lower cost of capital in debt markets (i.e., if a greenium exists). I compare the cost of capital for green and brown automobile asset-backed securities (auto ABS). Auto ABS are highly standardized debt instruments which securitize pools of automobile loans, collateralized by vehicles. Senior tranches of auto ABS have safe-asset status similar to US treasuries (Gorton, 2017). Only a few parameters distinguish auto ABS deals from each other besides their collateral pool. However, the carbon dioxide (CO_2) emissions of collateral pools, an objective measure of *greenness*, vary substantially. My identification strategy allows me to answer the following question: once an investor can be sure that the loan they extend will be paid back at the agreed time, do the CO_2 emissions of the collateral influence their choice to invest – and is this choice reflected in prices?

My framework allows me to rule out potential confounders that affect both cash flow and ESG-criteria in an effort to identify causal effects. Two key features of the securitization market allow me to do this. First, the security design of auto ABS reduces the number of risk factors. Due to time and risk tranching[2] and other credit enhancements (e.g., reserve accounts, over-collateralization, and excess spreads), the main risk factor that AAA-rated senior tranches of auto

[1] USSIF (2022): Sustainable Investing – Money Managers 2022

[2] Senior tranches receive interest and amortization payments first while defaults are borne first by junior tranches.

ABS deals are exposed to is prepayment. Second, the likelihood of early repayment is determined by borrower and loan characteristics and is not a characteristic of the collateral – while *greenness* is. The distinction between borrower and collateral characteristics together with granular loan-level data allows me to causally disentangle cash flow from ESG considerations.

I collect loan-level data on the collateral pools of 278 public auto ABS deals by 21 issuers from 2017 to 2022. The loan-level data represents the collateral pools of all public auto ABS and consists of 18.8 million loans. The data contain a range of information on the borrower, contractual terms, and collateral associated with each loan. By merging data about the collateral at the make, model, and year level to data on CO_2 emissions from the Environmental Protection Agency (EPA), I am able to estimate the lifetime emissions of each vehicle in a collateral pool.

The empirical tests I perform relate the emissions of collateral pools to issuance prices of AAA-rated senior tranches of auto ABS while controlling for the determinants of prepayment risk. The granularity of the loan-level data allows me to control for ex-post prepayment. Controlling for *perfect foresight* at issuance together with high-dimensional fixed effects removes as much unobserved heterogeneity as possible.[3] To test if ESG preferences are present in quantity data, I turn to portfolio holdings of mutual funds. If investors with ESG preferences are infra-marginal in setting prices in the auto ABS market, one can still learn about their preferences by observing their quantity choice using a revealed preference argument.

I first document the large cross-sectional dispersion of CO_2 emissions. Auto ABS issued by Honda's captive lending arm finance an average of 45 tons (t) of CO_2 per vehicle, while deals issued by Ford Motor Credit finance an average of 95t per vehicle. I show that the amount of CO_2 emissions and thus the greenness of an auto ABS deal can be measured by the share of trucks in its collateral pool. A 1 percentage point increase in the truck share increases the amount of financed CO_2 by 1.02t per vehicle. This sufficient statistic is available to investors in the prospectus and enables them to distinguish green from brown securities.

Using the cross-sectional dispersion in CO_2 emissions, I show that the existence of a greenium depends on whether it is possible or not to account for observed and unobserved heterogeneity across securities. In a *naive* specification that only accounts for market conditions at issuance, my results show a statistically significant greenium of around 14bps for auto ABS issuances with emissions in the lowest quartile. However, this greenium shrinks towards zero and flips sign when one accounts for differences in the security design of auto ABS (e.g., different assumed prepayment speeds). This highlights the importance of being able to account for differences in

[3] One interpretation of the perfect foresight variables is that they approximate prepayment models that investors use to price this risk.

security design. Moreover, after controlling for ex-post realizations of prepayment risk, a statistically significant "brownium" emerges. The brownium indicates that *higher*-emission collateral pools have a lower cost of capital. Going from the 20th percentile of CO_2 emissions intensity to the 80th percentile (a 1.85 standard deviation distance equivalent to comparing Toyota to Ford issuances) induces a brownium of 5bps. The magnitude of this brownium is economically meaningful, as the median issuance spread is around 32bps. This brownium is robust to alternative specifications, measures of greenness, and samples.

My results show that ESG preferences are not detectable in price data. Asset pricing theory thus suggest that ESG investors are not marginal in setting prices. If ESG investors are infra-marginal in setting prices, their preferences for greener assets would still be reflected in greater quantities of greener assets. In seminal work, Hong and Kacperczyk (2009) argue that norm-constrained investors, like endowments or ESG mutual funds, are more likely to invest in responsible securities compared to unconstrained investors. To investigate if the CO_2 emissions of the collateral influence ESG investors' decision to invest in auto ABS, I turn to portfolio data of mutual funds. Mutual funds are key investors in the auto ABS market. An average of 23% of issuance size per deal is held on their balance sheets. For some senior tranches the share of issuance size that is held by mutual funds can go as high as 85%. I identify 143 dedicated ESG-bond mutual funds. Out of these 143 funds, 35 hold at least one position in auto ABS, and 32 hold at least two positions.

Two surprising facts emerge: ESG funds are (i) invested across the full distribution of CO_2 emissions and (ii) tend to have larger portfolio shares in higher-emission deals compared to non-ESG funds. ESG funds allocate approximately 32% less capital to auto ABS in the bottom quartile of emissions. These two findings are hard to reconcile with common ESG strategies. There is an active debate about which strategy ESG investors should follow: *exit* (divestment or exclusion) versus *voice* (shareholder activism).[4] Investments in debt securities are restricted to exit since creditors have no voting rights.[5] The most popular exit strategies are outright exclusion or best-in-class investment. However, inconsistent with these strategies, ESG mutual funds invest across the full distribution of CO_2 emissions and allocate more capital towards higher-emissions deals compared to non-ESG funds.

The main contribution of this paper is demonstrating the absence of ESG preferences in both price and quantity data. I innovate in two ways over the existing literature. First, focusing on

[4] Broccardo et al. (2022) analyze the relative effectiveness of exit and voice strategies. Edmans et al. (2022) analyze whether exclusion or best-in-class investment is more effective.

[5] This is especially the case for auto ABS since these securities are issued by special purpose vehicles which are bankruptcy-remote from their sponsoring entity.

senior tranches of auto ABS allows me to isolate the effect of greenness on prices and quantities while holding traditional risk factors like default and prepayment risk constant across securities. Holding risk factors which influence expected cash flows constant is important as prices potentially react to greenness-induced risk. For example, Seltzer et al. (2022) show that corporate bonds with poor environmental profiles have higher yield spreads, particularly when these corporations' facilities are located in states with strict regulatory enforcement. Regulatory risk caused by poor environmental performance is distinct from ESG preferences affecting prices. For senior tranches of auto ABS, borrower and loan characteristics determine cash flow risk while a single collateral characteristic, CO_2 emissions, determines ESG friendliness. Separating greenness from cash flow risk allows causal inference about the effect of ESG preferences on prices and quantities, previously obscured by confounding factors like regulatory risk. Second, I use *continuous* measures of greenness rather than a discrete definition. Previous studies of the greenium in debt markets have focused on municipal and corporate bonds (e.g., Zerbib, 2019, Baker, Bergstresser, Serafeim and Wurgler, 2022, Larcker and Watts, 2020). These bonds are either green if they fund "environmentally friendly projects'' (e.g., solar energy production or waste water treatment) or not green. There can be qualitative and quantitative differences between green projects. Economists do not know yet how ESG investors react to these differences. I address this problem by focusing on a single, continuous dimension of environmental relevance. Note that auto ABS are inherently brown since they finance CO_2 emissions. There are, however, large cross-sectional differences in emissions, making some auto ABS deals greener than others.

To summarize, this paper shows in a causal setting that CO_2 emissions, an objective measure of greenness, are not priced in debt markets despite the rise of ESG investing over the past decade. By looking at quantities, the paper further shows that mutual funds marketed as ESG (i) are invested across the full distribution of CO_2 emissions and (ii) tend to hold higher portfolio shares in high-emissions issues compared to non-ESG funds. Both quantity findings are surprising since greener auto ABS are not riskier and browner securities do not pay higher yields. This investment behavior is hard to reconcile with common ESG strategies and has implications for retail investors of ESG mutual funds, fund managers, and policy makers. The results suggest that ESG mutual funds are not investing in the most environmentally-friendly securities, which raises questions about the effectiveness of ESG strategies.

The rest of the paper is organized as follows. The remainder of the introduction discusses the related literature. Section 2 provides a brief overview over the ABS market for auto loans and establishes stylized facts about the emissions content of auto ABS. Section 3 describes the data. Section 4 discusses the identification strategy. Section 5 reports estimates of the greenium and

discusses results. Section 6 studies ESG preferences in the portfolio holdings of mutual funds. Section 7 concludes.

Related Literature The rise of ESG investing has sparked an active literature.[6] Theoretical contributions (e.g., Heinkel et al., 2001) have highlighted that if the share of ESG conscious investors is large enough, green firms will have a lower cost of capital.

One of the earliest theoretical contribution on the equilibrium impact of investors with "green" or ESG preferences is Heinkel et al. (2001). The paper models an equilibrium where the investment behavior of ESG investors raises the cost of capital of polluting firms and lowers the cost of capital for green firms. Pástor et al. (2021) propose a theoretical framework to model the impact of changes in sustainability preferences on asset prices. The authors analyze the equilibrium implications of sustainable investing and conduct an analysis of welfare and social impact. Oehmke and Opp (2020) characterize conditions under which socially responsible investors impact firm behavior, in a setting in which firm production generates social costs and is subject to financing constraints. Berk and van Binsbergen (2021) study the theoretical effect of divestment in a single period mean-variance environment. They calibrate their model to current data and demonstrate that the impact of ESG investors on firm's cost of capital is small. One implication of their paper is that investors are likely to be more effective in reducing social costs by investing, rather than divesting, in socially costly firms and using their rights of control to alter corporate policy. Feldhütter and Pedersen (2022) study theoretically whether capital structure is irrelevant for enterprise value and investment when investors care about ESG issues. If markets are perfect and ESG is additive an ESG-modified Modigliani-Miller theorem holds in which green bonds do not lower the cost of capital.

The result of this paper, that differences in greenness are not priced or even induce a brownium, stands in contrast to prior empirical results in the literature. For example, Pástor et al. (2022) document a near-arbitrage opportunity in which German government green "twin" bonds relative to their non-green twins trade on average at 5bps lower yields. Baker, Bergstresser, Serafeim and Wurgler (2022) estimate a greenium of about 6bps in a sample of over 2,000 U.S. municipal and corporate green bonds, Zerbib (2019) estimates a greenium of 2bps in a sample of over 1,000 supranational, sub-sovereign and agency, municipal, corporate, financial, and covered green bonds. In contrast, and similar to the findings of the present study, Larcker and Watts (2020) find no premium on municipal green bonds. The authors point out shortcomings in the estimation methods used in previous studies and rule out several plausible alternative explanations

[6] Both Gillan et al. (2021) and Hong and Shore (2022) provide recent literature reviews.

for the lack of an observed premium.

The results of the present study and the ostensibly conflicting results from the literature can could be reconciled by a "green window dressing" effect where securities that are especially marketed as green trade at a premium while securities that markedly differ in their true greenness but are not marketed as such do not. I leave the exploration of this channel for future research.

I also contribute to the literature on auto ABS. The literature on auto ABS is relatively thin compared to other consumer ABS such as mortgage-backed securities. Klee and Shin (2020) study asymmetric information and signaling in post-crisis offerings in the auto ABS market using the same data as the present study. Benetton et al. (2021) analyzes European auto ABS data and finds that the vertical integration of manufacturing and credit provision allows captive auto lenders to increase cash collected from car sales via a credit fire sale channel.

2 Securitized Auto Loans and their Emissions

This section gives a brief introduction of the market for securitized auto loans, explains key concepts, and establishes several stylized facts about the CO_2 content of this type of security. It also shows that a simple summary statistic, the share of trucks, is enough to differentiate green from brown deals.

2.1 The ABCs of Auto ABS

Lending for automobiles has been one of the most commonly used forms of consumer credit, as automakers find it advantageous to issue loans to increase sales. Auto ABS were among the first consumer ABS to come to the market in the 1980s, when securitization of consumer loans grew popular. At the end of 2021, outstanding auto ABS accounted for around 18% of total outstanding auto loans in the US, with a value of roughly $220 billion.[7] There are on average 10 new public auto ABS issuances with a total notional value of approximately $13 billion per quarter over my sample period from 2017 to 2022. Compared to corporate and municipal bond markets, the security design of the auto ABS market is highly standardized. Only a few parameters distinguish auto ABS deals from each other besides their collateral pool. These are the weighted average life of a tranche, credit spread over a yield curve, assumed prepayment speed, and an option to call the bond prematurely.

Due to the high level of standardization in the auto ABS market and the safety of these se-

[7] see SIFMA, U.S. ABS issuance and outstanding.

curities, auto ABS are highly liquid. Auto ABS have similar or lowers levels of liquidity and market risk than agency mortgage backed securities that trade in the $200 billion daily volume TBA market (see He and Mizrach, 2017).

All of the 278 deals in my sample are structured as monthly amortizing with higher seniority tranches receiving repayments first. Appendix Figure A1 shows examples of typical auto ABS deal structures from tear sheets sent to investors. Auto ABS are typically priced as a spread over a credit curve. These curves can differ by tranche. Senior tranches are most commonly priced against the Eurodollar Synthetic Forward Curve (EDSF), but have recently shifted to pricing against the Interpolated Yield Curve (I-Curve) extracted from on-the-run Treasuries. For a description of the corporate bond underwriting process by investment banks, which is similar to the ABS underwriting process, see Siani (2022). The auto ABS market can be split into prime and sub-prime shelves based on the creditworthiness of their underlying loans; sub-prime shelves command higher issuance spreads.

Given time and risk tranching, high levels of over-collateralization, and other credit enhancements (e.g., reserve accounts or excess spreads), prepayment is the main risk factor for investors in senior tranches of auto ABS. Prepayment risk can materialize in several ways, primarily as the early repayment of loans by consumers but also if a borrower defaults on a loan and the car is repossessed. This risk is typically measured and priced using a prepayment model.[8] Auto ABS deals are also characterized by embedded clean-up call options. These options allow the issuer to prematurely call the outstanding notes if the remaining pool balance drops below a certain percentage (most commonly 5% or 10%). The clean-up call option, however, is not relevant for the most senior tranches as those have a short weighted average life (WAL)[9] and are paid off before the pool-balance reaches the cutoff point at which the option is in-the-money for the issuer. While all of the auto ABS deals analyzed here are registered securities and publicly traded, there is a considerable amount of auto ABS which are privately placed under Regulation S and Rule 144A. Private placements, however, do not need to publicly disclose loan-level information and are therefore excluded from my sample.

[8]Prepayments on auto ABS can be measured relative to a prepayment standard or model. The model used in all prospectuses of the 278 deals in my sample is the Absolute Prepayment Model (APS). APS represents an assumed rate of prepayment each month relative to the original number of receivables. APS further assumes that all of the receivables are the same size, amortize at the same rate, and that each receivable in each month of its life will either be paid as scheduled or be prepaid in full. For example, in a pool of receivables originally containing 10,000 receivables, a 1% APS rate indicates that 100 receivables are expected to be prepaid each month.

[9]The WAL is the average length of time that a dollar of unpaid principal on a loan, a mortgage, or an amortizing bond remains outstanding.

2.2 Stylized Facts about CO2 Emissions in auto ABS

This empirical analysis studies the effect of CO_2 emissions of the vehicles that serve as collateral for asset-backed securities on issuance prices and quantities purchased by ESG investors. Passenger vehicles are responsible for approximately 17% of total US CO_2 emissions and the negative externalities associated with GHG emissions of motor vehicles are widely accepted and can be measured. The CO_2 emissions of collateral pools are predetermined by the past technological choices of manufacturers and vehicle demand which alleviates concerns about greenwashing in the design of the securities.

Publicly traded auto ABS need to disclose loan-level data on the collateral pool. A unique feature of this data is that it allows calculations of the financed CO_2 emissions for each ABS's collateral pool. That is, the amount of expected emissions that the auto ABS collateral pool b is financing is given by the sum over each vehicle i as:

$$\mathbb{E}_t\left[\text{Financed } CO_2 \text{ Emissions}\right]_b = \sum_{i \in b} \underbrace{CO_2 \text{ Emissions}_i \times \mathbb{E}_t\left[\text{Survival-Weighted Miles}\right]_i}_{\text{Expected Emissions}} \times \underbrace{LTV_{0,i} \times \text{Outstanding Balance Share}_{i,t}}_{\text{Financing Adjustment}}. \quad (1)$$

The first term of the right-hand side of eq. (1) is the CO_2 emissions for a given vehicle i measured in tons of CO_2 per mile driven. The second term is the expected survival-weighted vehicle miles traveled over the lifetime of the vehicle. The product of these terms is the total expected lifetime emissions of a new car.[10] The loan-to-value (LTV) part of the financing adjustment of eq. (1) reflects the fact that not all of the expected CO_2 emissions are directly financed through a loan since many consumers make down-payments at the time of purchase. The financing adjustment further takes into account that many loans in bond-pools are warehoused with the original lender for a some time and only securitized after a certain balance has been paid off.

By matching the vehicle collateral of auto ABS deals to their respective emissions data from the EPA, I am able to calculate the expected lifetime emissions for each vehicle financed. The CO_2 emissions of each ABS deal vary substantially across and within issuers. Figure 1 shows boxplots of the average financed CO_2 emissions (in t) across all ABS collateral pools for issuances by Honda, Toyota, Santander Bank, General Motors (GM), and Ford. The figure illustrates a striking fact: similar to a motorist's choice of a high-emissions vehicle (e.g., Ford F-150) versus

[10] The survival-weighted vehicle miles traveled of used vehicles is adjusted to reflect the remaining expected lifetime of the vehicle.

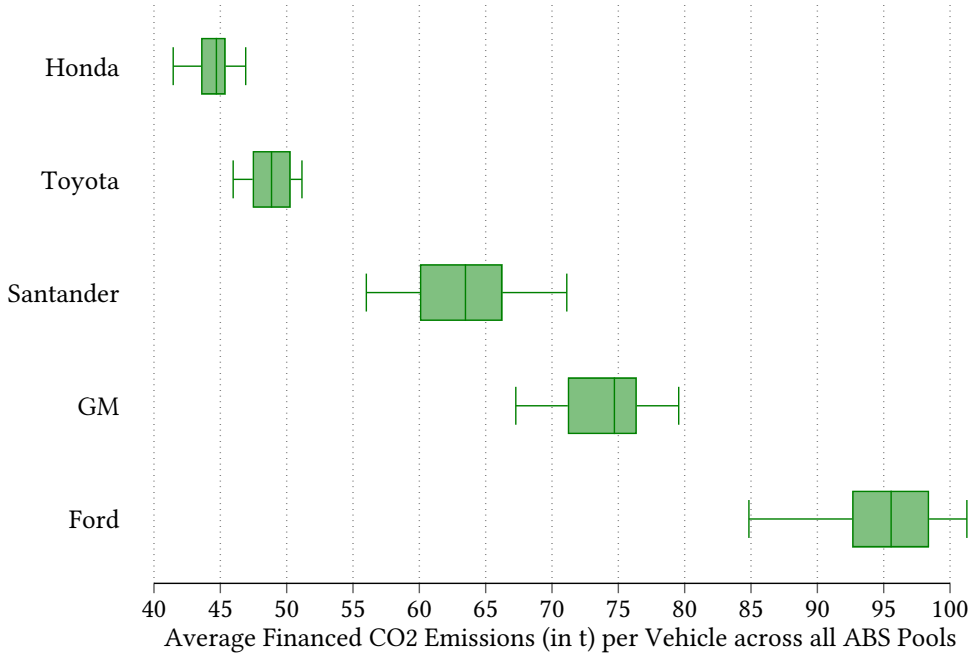


Figure 1: Dispersion of CO_2 Emissions across all ABS Pools for select issuers

a low-emissions vehicle (e.g., Toyota Prius), an investor has the choice to invest in an auto ABS which finances high-emissions vehicles (e.g., Ford) versus an auto ABS which finances relatively low-emissions vehicles (e.g., Toyota).

The sizable differences in emissions across ABS deals are largely explained by the vehicle type composition of the collateral pool. For example, across all 21 ABS deals issued by Honda, the average share of passenger vehicles in the collateral pools is 40.2%, the average share of SUVs is 58.3%, and the average share of trucks is 1.5%. In contrast, across all 16 ABS deals issued by Ford, these shares are 11.8%, 38.7%, and 49.5%, respectively. The vehicle type composition for captive auto lenders is related to the product offerings by their vehicle manufacturing parents and consumer demand for these products. The high share of trucks and the associated high-emissions of Ford's auto ABS reflect the fact that Americans have bought more F-Series trucks than any other vehicle for more than four decades.[11]

Environmentally conscious investors can use the fact that differences in emissions across ABS deals are explained by the vehicle type composition to distinguish brown from relatively green auto ABS. Table 1 reports coefficients from a regression of average tCO_2 emissions per vehicle on the share of trucks in the collateral pool. For each 1 percentage point increase in the truck share, the average amount of CO_2 per vehicle increases by 1.021 ton. The different vehicle type

[11] Kelley Blue Book: Ford F-150 Retakes Best-Selling Vehicle Crown

Table 1: Regression of Mean tCO_2 Emissions per Vehicle on Truck/SUV Shares

	Constant	Truck share	SUV share	Adj. R^2	N	Mean Dep. Variable
β	44.049***	1.021***	0.209***	0.746	278	70.607
(se) or sd	(1.892)	(0.036)	(0.040)			15.605

Dependent variable: Mean tCO_2 emission per vehicle of collateral pool.

shares are reported in the securities prospectus and are readily available as a variable in the loan level data. This presence of a simple summary statistic which gives an accurate reading of the CO_2 intensity implies that auto ABS investors have a "sufficient statistic" in their information set to distinguish brown from relatively green auto ABS.

3 Data

This section details information about the loan-level data used to construct the measures of relative greenness of auto ABS. This section also provides information about the issuance level data used in the empirical tests.

3.1 Loan-level Data

The primary data on loan-level information about borrowers and vehicles comes from the Securities and Exchange Commission (SEC) form ABS-EE. Form ABS-EE is associated with post-financial crisis reporting requirements under Regulation AB, which went into effect on November 23, 2016. Under the reporting requirement, all prospectuses for public offerings of asset-backed securities must also submit accompanying loan-level information in electronic format. Further, every month loan-level information about the performance of the loan pool are to be provided.

The loan-level data covers all collateral pools of public auto ABS issued from January 2017 to December 2022, and consist of approximately 18.8 million unique loans and 278 ABS deals issued by 21 issuers.[12] The data contains a range of information on the originator, borrower, and collateral associated with each loan. Appendix Table A1 presents summary statistics of loan-level

[12]These issuers are: Ally Financial, AmeriCredit, BMW Financial, Capital One Bank, CarMax, Carvana, Exeter Finance, Fifth Third Bank, Ford Credit, GM Financial, Honda Finance, Hyundai Capital, JM Family, Mechanics Bank, Mercedes-Benz Financial Services, Nissan Finance, Santander Bank (DRIVE), Santander Bank (SDART), Toyota Motor Credit, USAA Federal Savings Bank, and Volkswagen Credit. I distinguish Santander's sub-prime shelf SDART from their deep sub-prime shelf DRIVE since the latter carries higher (prepayment) risk in the form of defaults. Similarly, I distinguish GM's subprime shelf AmeriCredit from their prime shelf GMCAR.

data. The average borrower in my sample finances roughly $26,000, at 90% loan-to-value, at an 7.6% interest rate for 67 months. Their credit score is 712 and their monthly payment to income ratio is 0.08. The vehicle the average borrower is financing is worth $27,733.

3.2 Emissions Data

The data on CO_2 emissions comes from the Environmental Protection Agency (EPA). This data is matched on the make, model, and model year to the loan-level collateral data. Estimates of survival-weighted vehicle miles traveled (SVM) by vehicle type are pulled from the EPA CAFE standard emissions simulator. The SVM estimates vary by vehicle type and assume up to 40 years of useful life, with the bulk of miles driven in the first 10 years (e.g., for passenger cars, around 70% SVM accrue in the first 10 years). The EPA estimates that trucks and SUVs are driven at a higher yearly mileage and have a higher survival probability per year than sedans and compact cars. Appendix Figure A2 plots these SVM curves by vehicle type.

The average vehicle in my sample will be driven for 202,963 miles, of which 162,450 miles are indirectly financed by the proceeds of the bonds. Over its total lifetime, the vehicle will emit approximately 79t of CO_2, of which 63t are financed by ABS proceeds.

There is, however, considerable heterogeneity among emissions of the financed collateral since the sample includes fully-electric vehicles, sedans and compact cars, SUVs, as well as pick-up trucks and other high-emissions vehicles. The 10 most popular vehicles in my sample demonstrate this heterogeneity. These are, in order, Toyota Camry (sedan, on average 60t of CO_2 emissions over full lifetime), Toyota RAV4 (SUV, 73t), Toyota Corolla (sedan, 53t), Nissan Rogue (SUV, 62t), Chevrolet Silverado (truck, 120t), Honda Civic (sedan, 51t), Nissan Altima (sedan, 59t), Honda CR-V (SUV, 65t), Honda Accord (sedan, 62t), and Ford F-150 (truck, 114t).[13]

3.3 ABS Deal Data

I collect information about the structure of each deal from their respective prospectuses filed with the SEC (Form 424H, 424B5, and FWP). The data contains information about the characteristics of each deal and its associated tranches, e.g., issue date, ratings, coupon, spreads, issuance amounts, weighted average life (WAL), and associated book running banks. Issuance spreads are calculated as the difference between the issuance yields and yield curve estimates of Filipović et al. (2022) for which the maturity is matched to the WAL of a given tranche.

[13] These top 10 models make up roughly 28% of observations and closely track the lists of most sold cars in the US over the past 5 years.

Table 2: Summary Statistics Issuance-Level Data (A-2 Tranche)

	Mean	SD	Median	Min	Max	N
Total Deal Size ($ m)	1,239.33	343.38	1,250.00	367.31	2,663.82	278
A-2 Tranche Size ($ m)	367.24	132.48	362.50	42.40	746.94	278
Weighted Average Life (Years)	0.98	0.32	1.01	0.37	3.50	278
Spread (bps)	42.88	36.29	32.43	6.13	407.91	278
Coupon (%)	1.92	1.30	1.86	0.14	5.81	278
Subprime ABS	0.27	0.45	0.00	0.00	1.00	278
Captive Lender	0.37	0.48	0.00	0.00	1.00	278
Number of Loans	67,335.08	25,365.61	66,114.00	15,212.00	180,352.00	278
Warehousing Time (Months)	9.55	4.40	9.18	1.33	21.07	278
Income Verified	0.12	0.26	0.01	0.00	1.00	278
Used Vehicles	0.42	0.34	0.30	0.02	1.00	278
Mean Credit Score	706.74	75.07	738.93	564.98	788.46	278
Expected tCO2 per $100,000	293.55	51.22	296.69	161.51	456.16	278
Avg. Exp. tCO2 per Vehicle	70.61	15.60	67.72	42.94	125.73	278
Financed tCO2 per $100,000	219.82	40.21	211.31	107.10	311.78	278
Avg. Fin. tCO2 per Vehicle	58.03	12.83	54.46	40.54	101.27	278

Table 2 presents summary statistics for the A-2 tranches of each deal. The average deal size is $1.2bn, of which, on average, 25% are associated with the A-2 tranche. The average A-2 note is issued with a spread of 43 basis points and a WAL of roughly one year. Around 42% of deals are issued by captive lenders and around 25% are sub-prime issues. The average deal collateralizes around 67,335 vehicles, of which approximately 42% are used. An investor who buys $100,000 of the average A-2 tranche finances approximately 220t of CO_2 over the remaining life of the collateral. The average car an investor finances will emit approximately 58t of CO_2 over the remaining life of the collateral.

4 Identification Strategy

While standard no-arbitrage economic theory suggests that identical brown and green securities should be priced identically, there is a growing literature that argues otherwise. Both theoretical (e.g. Heinkel et al., 2001) and empirical contributions (e.g. Pástor et al., 2022, Baker, Bergstresser, Serafeim and Wurgler, 2022, Zerbib, 2019) show effects of non-pecuniary preferences on asset prices in equilibrium. Studying the effect of non-pecuniary preferences on asset prices is fundamentally challenging. Unobserved heterogeneity could create spurious correlation between greenness and returns which complicated causal identification.

Issues related to unobserved heterogeneity are mitigated in the case of auto ABS. The security-

design of senior auto ABS practically ensures that the only major risk-factor is prepayment risk.[14] Moreover, prepayment risk is a borrower characteristic and not a characteristic of the underlying collateral which induces the relative greenness. As the greenness of a given security is often subjective, previous studies have needed to rely on noisy and discrete measures. In auto ABS, however, an objective measure of relative greenness is given by the CO_2 emissions of the collateral pools.

I test if non-pecuniary preferences induce a greenium using the following specification:

$$\text{Issuance Spread}_{obt} = \alpha \text{Green}_b + \zeta' X_{obt} + \gamma_t + \varepsilon_{obt} \tag{2}$$

for bond tranche b issued by originator o issued at time t. Green_b is an indicator variable equal to one if the collateral pool's CO_2 emissions per \$100,000 principal are below the 25th percentile of all auto ABS deals and zero otherwise. The coefficient of interest is α, which reflects the greenium investors are willing to accept for a relatively more green security. While the specification in eq. (2) using an indicator variable is consistent with the existing literature on the greenium, there is little ex-ante reason to believe that markets only attach a greenium to the lowest quartile of emissions intensity. I therefore additionally run the following specification to test if there are linear effects:

$$\text{Issuance Spread}_{obt} = \beta \left(\mathbb{E}_t \left[\text{CO}_2 \text{ Emissions}_b \right] \right) + \zeta' X_{obt} + \gamma_t + \varepsilon_{obt} \tag{3}$$

The coefficient of interest is β, which reflects the sensitivity of issuance spreads to CO_2 emissions of the underlying collateral pool. I operationalize the expected emissions of a given collateral pool, $\mathbb{E}_t \left[\text{CO}_2 \text{ Emissions}_b \right]$, in two ways: (i) as tCO_2 per \$100,000 principal and (ii) as an average of tCO_2 per vehicle.

The identifying assumption in both cases is that CO_2 emissions are uncorrelated with the error term conditional on traditional risk factors. That is, once prepayment and default risk are accounted for, assignment of CO_2 emissions are "as good as random".

The identification assumption would be violated if vehicles with higher CO_2 emissions were to carry higher default or prepayment risk independently of borrower or loan characteristics. For example, if high-emissions vehicles are exposed to *transition risk* we would expect an downward (upward) bias in α (β). High-emissions vehicles could be exposed to transition risk if, e.g., markets believe that the government will soon implement a CO_2 tax, raise gas taxes, or outlaw high-

[14] Liquidity and market risk are of second order as the low bid-ask-spreads of \$0.04 and immediate market impact estimates of \$0.01 indicate. For comparison, this is the same bid-ask-spread and lower immediate market impact as in the \$200bn daily volume TBA market for agency MBS (see He and Mizrach, 2017).

emissions vehicles, which in turn makes borrowers default on their loan. The security-design of auto ABS helps shed light on the role of transition risk, since subordinate tranches have longer maturities and by design carry higher exposure to risk, including transition risk. Another possible source of unobserved heterogeneity in auto ABS would be a strong home-bias for American car manufacturers like Ford or GM. In this case, one would expect a upward (downward) bias in α (β), caused by the relatively stronger demand for manufacturers of high-emissions vehicles, which happen to be American. I directly control for a differential demand impact of a potential home-bias by including a US-issuer fixed effect in the regression specifications. The specifications further control for subprime issue fixed effects, captive lender fixed effects, as well as fixed effects relating to the pricing speeds of the different deals, i.e., clean-up call fixed effects and assumed prepayment speed fixed effects.

I further abate the possibility that CO_2 emissions relate to unobserved aspects of the credit and collateral quality in two ways. First, since issuers have to file monthly updates on the loan-level data, one can create perfect foresight variables that account for future realizations of prepayment risk. The vector X_{obt} includes the average realized difference of monthly absolute prepayment speed (APS) to its prospectus assumption and the average realized percentage of loans which are more than 30 days delinquent. These variables capture unobserved heterogeneity in spreads which directly relate to voluntary and involuntary prepayment risk. Second, the main specification focuses on the second-most senior A-2 tranches of each deal.[15] All of the A-2 tranches are AAA-rated due to their time and risk tranching. Time tranching ensures that the A-2 tranches have a weighted average life of only one year. The average attachment point[16] of these tranches is around 50% because of risk tranching. Note that if a borrower defaults on their loan, the car usually gets repossessed and sold again. To holders of senior tranches the default of an individual borrower, repossession of the collateral, and subsequent recovery is a single "involuntary" prepayment event. The difference in outstanding loan balance and recovery value, i.e. the haircut, is born by the most junior tranche.[17] The high levels of credit and time tranching imply that any alternative explanation that posit a risk-factor that is particularly associated with high emission collateral, would need to be (a) very large in size and (b) have a high chance to materialize within one year. Moreover, this risk should be priced more strongly in subordinate tranches – a testable implication.

[15] The most senior A-1 tranches are notes with a weighted average life of less than three months; they are similar in character to short term commercial paper. A-1 notes are often retained by the issuer or privately pre-placed.

[16] The attachment point indicates the minimum percentage of pool-level losses at which a given tranche begins to suffer losses on its principal.

[17] The historic recovery values are around 60% and 45% for prime and subprime loans, respectively. (Structured Finance Association: Trends in Auto ABS).

Taken together, these two points: (i) the availability of perfect foresight variables that account for prepayment risk and (ii) the short-term, safe nature of the securities make it unlikely that traditional risk factors explain any finding. The recovered estimate of α and β therefore likely reflects the market's taste for relatively greener deals.

To improve the precision of the estimates I further include a host of control variables. The vector X_{obt} contains controls for the current market conditions, tranche characteristics, and collateral pool characteristics. To control for current market conditions, X_{obt} includes estimates of the high-quality corporate yield curve for maturities of 1, 6, 12, and 48 months, the level of the VIX, the standard deviation of the VIX during the month before issuance, and the level of the 5-year breakeven inflation rate at issuance. The included tranche characteristics are the weighted average life and the default attachment point. The vector X_{obt} also contains collateral pool controls that are potentially indicative of prepayment and default risk. These are the collateral pool averages of LTV, credit scores, used vehicle share, remaining loan balance share, interest rate, remaining term, and share of loans with income verification. Collateral pool variables are weighted by outstanding loan amount. Collateral pool and tranche characteristic have individual slopes across subprime and prime issues to account for potentially different pricing relevance of those variables. Standard errors are clustered at the year-month level to account for common variation among spreads caused by current market conditions.

Appendix Table A2 shows a covariate balance table. Other than financed tCO_2 emissions per \$100,000 (i.e., the sorting variable), only the average credit score and 5 year breakeven inflation of green and brown deals have statistically different means at the 5% level. The differences in means, however, are less than one standard deviation. This shows that relatively green deals do not differ substantially along observable dimensions which influence the pricing of auto abs deals. This is important, since it implies that ESG investors have the choice between green and brown while holding observable risk-characteristics constant.

5 Results

This section shows the results of the tests and identification strategy discussed in Section 4. The section also provides an overview over the robustness check. Lastly, it discusses the results and puts them into context of the literature of the greenium.

5.1 Estimates of the Greenium in auto ABS

The identification strategy allows me to answer the following questions: once an investor can be sure that their loan will be paid back at the agreed time, do the CO_2 emissions of the collateral influence their choice to invest – and is this choice reflected in price? An $\alpha < 0$ (or $\beta > 0$) would suggest that they prefer relatively greener collateral., i.e., the presence of non-pecuniary preferences in the market.

Panel A and Panel B of Table 3 present estimates of the greenium using the pricing models of equation (2) and equation (3), respectively. The specification in column (1) controls only for current market conditions as well as year-month FE. The greenium of an auto ABS deal with emission-intensity in the bottom 25% of issuances is estimated to be 14.40 basis points and statistically significant ($t = 5.78$). Similarly, Panel B column (1) shows that a one standard deviation higher CO_2 emissions intensity is associated with 0.143 standard deviation (sd) higher issuance spread, indicating the presence of a greenium of 5.19bps per sd CO_2 emissions ($\approx 0.143 \times 36.29$bps).

This naive specification, however, fails to control for observed heterogeneity. Column (2) adds tranche characteristics to control for heterogeneity in the design of the securities. These are the attachment point, weighted average life, and fixed effects for the assumed prepayment speed (APS) and clean-up call option. The estimated greenium shrinks towards zero, flips sign, and becomes statistically indistinguishable from zero. Column (3) further adds subprime and captive lender fixed effects as well as prepayment speed predictors like the mean credit score with the effect of further shrinking the estimates towards zero.

Column (4) introduces two perfect foresight variables. These are the difference of the average realized APS to the assumed APS from the prospectus and the realized average share of loans that are delinquent (30-60 days). Adding these variables increases the R^2 by around 23%.[18] Moreover, in Panel B the sensitivity of issuance spreads with respect to CO_2 emissions becomes statistically significant ($t = 2.81$). The coefficient implies that a one standard deviation higher CO_2 emissions intensity is associated with 0.08 sd *lower* issuance spread, indicating a *brownium* of about 3bps per sd CO_2 emissions. Going from the 20th percentile (189t) of CO_2 emissions intensity to the 80th percentile (262t), a 1.85sd distance which is equivalent to comparing Toyota to Ford issuances, induces a brownium of 5bps ($\approx 0.08 \times 1.85 \times 36.29$bps). The magnitude of this brownium is economically meaningful as the median issuance spread of all A-2 tranches is around 32bps.

[18] Interestingly, the A-2 tranches with higher realized prepayment speeds have lower issuance spreads. Note, however, that this is an ex-post finding and does not necessarily imply ex-ante mispricing. For example, ex-post we only observe one realized path of prepayments but ex-ante a continuum of paths are possible. The probability distribution over these possible paths needs to be priced ex-ante.

Table 3: Estimates of the Greenium in the auto ABS market

Panel A: Green Indicator

	(1) Issuance Spread	(2) Issuance Spread	(3) Issuance Spread	(4) Issuance Spread
Green (tCO2<p25)	-14.40***	3.561	5.070	1.381
	(2.491)	(6.750)	(3.506)	(1.828)
Delta Realized ABS				-18.04***
				(4.229)
Realized % Loans Delinquent 30-60 days				-2.407
				(1.784)
Year-Month FE	Yes	Yes	Yes	Yes
Market condition controls	Yes	Yes	Yes	Yes
Tranche characteristics	No	Yes	Yes	Yes
Subprime, captive FE	No	No	Yes	Yes
Add. prepayment speed controls	No	No	Yes	Yes
Adj. R^2	0.554	0.669	0.795	0.967
Observations	273	273	273	271

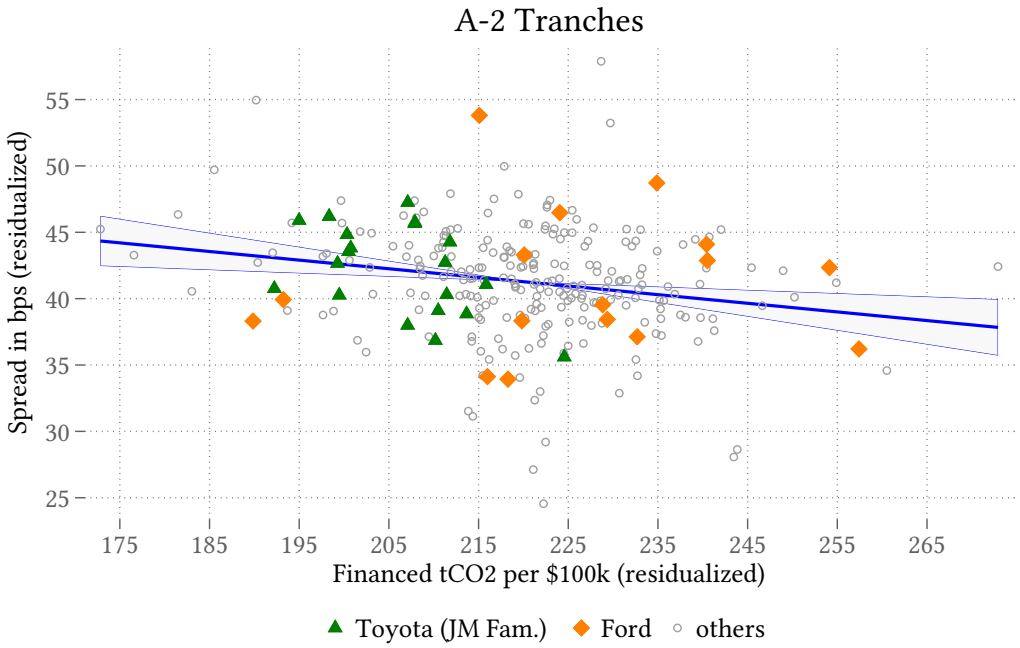
Standard errors in parentheses clustered at year-month (K=71).
+ p<0.10, * p<0.05, ** p<0.01, *** p<0.001

Panel B: Financed tCO_2 per $100,000

	(1) Issuance Spread	(2) Issuance Spread	(3) Issuance Spread	(4) Issuance Spread
Financed tCO2 per $100k	0.143+	-0.297	-0.172	-0.0755**
	(0.0735)	(0.237)	(0.111)	(0.0269)
Delta Realized ABS				-0.142***
				(0.0335)
Realized % Loans Delinquent 30-60 days				-0.197
				(0.138)
Year-Month FE	Yes	Yes	Yes	Yes
Market condition controls	Yes	Yes	Yes	Yes
Tranche characteristics	No	Yes	Yes	Yes
Subprime, captive FE	No	No	Yes	Yes
Add. prepayment speed controls	No	No	Yes	Yes
Adj. R^2	0.549	0.690	0.798	0.968
Observations	273	273	273	271

Standard errors in parentheses clustered at year-month (K=71). Coefficients are standardized to unit variances.
+ p<0.10, * p<0.05, ** p<0.01, *** p<0.001

(a) *Brownium* in A-2 Tranches



(b) *Brownium* in A-3 Tranches

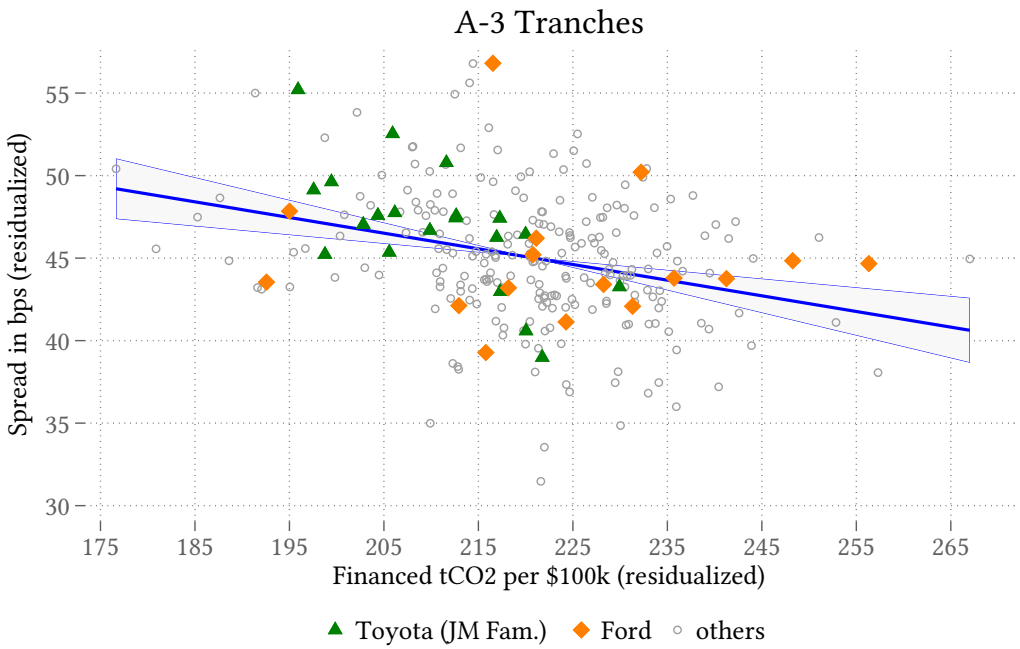


Figure 2: Residualized Scatter Plot of Issuance Spreads against tCO_2 Emissions

Figure 2 shows the slope of this brownium in a residualized scatter plot of issuance spreads against the amount of residualized tCO_2 emissions per \$100,000 for A-2 and A-3 tranches. The graphs depicts the specification of column (4) of Panel B in Table 3. In both tranches we see that collateral pools with higher-emissions get issued at lower spreads. The slope of this brownium is stronger in A-3 tranches than it is in the A-2 tranches of each deal.

5.2 Robustness

This subsection performs several robustness checks on the main result. It shows that the brownium result is robust to using alternative measures, tranches, and specifications.

Alternative Measures I perform a series of robustness test using different measures of relative greenness for each deal. These measures of relative greenness are (i) the average financed tCO_2 per vehicle, (ii) the average MPG of vehicles, (iii) the average truck share, and (iv) an independently constructed measure of greenness by the Kroll Bond Rating Agency (KBRA).[19] Note that the measures (i) to (iv) are not perfectly correlated ($0.1 < |\rho| < 0.47$) with the measure of financed tCO_2 per \$100,000 and therefore provide an independent signal about the relative greenness of each deal. While measures (i) to (iii) are highly correlated with the KBRA measure ($|\rho| > 0.75$), it serves as a good robustness check since it is independently constructed, publicly available, and a salient feature of the US car market. Table 4 shows that the results do not change qualitatively when using different measures of relative greenness.

Choice of Tranche So far, the analysis has used A-2 tranches as those all have similar characteristics across the different deals, have low credit risk, clean-up call options are non-binding, and have the highest observation count. However, the results are robust to the choice of any AAA rated tranche. Appendix Table A3 reports regression results across all AAA rated tranches. Note that the observation count varies as in some deals the A-1 and A-4 tranches are not offered publicly. Again the pattern emerges that if there is pricing it goes in the direction of a brownium higher emissions deals receiving lower spreads. The estimated brownium is even stronger in A-3 and A-4 tranches than in A-2 tranches.

[19] The KBRA released free reports in March 2021 (KBRA, 2021) and July 2022 (KBRA, 2022) mapping the EPA's vehicle greenhouse gas (GHG) scores to 247 auto ABS deals. The EPA's GHG scores range from 1 to 10 with higher values indicating lower emissions. The EPA's GHG scores are displayed on window labels attached to each new cars on display in dealer showrooms in the US since 2013.

Table 4: Estimates of the Greenium using different Definitions of relative *Greenness*

	(1) Issuance Spread	(2) Issuance Spread	(3) Issuance Spread	(4) Issuance Spread	(5) Issuance Spread	(6) Issuance Spread	(7) Issuance Spread	(8) Issuance Spread
Avg. Fin. tCO2 per Vehicle	-0.110^{+}	-0.0577*						
	(0.0571)	(0.0278)						
Avg. MPG			-0.166*	-0.0480^{+}				
			(0.0804)	(0.0245)				
Avg. Truck Share					-0.117	-0.0627^{+}		
					(0.0783)	(0.0358)		
Avg. GHG Rating (KBRA)							-0.0124	-0.0216
							(0.0292)	(0.0291)
Year-Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Market condition controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Tranche characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Subprime, captive FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Add. prepayment speed controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Perfect Foresight Controls	No	Yes	No	Yes	No	Yes	No	Yes
Adj. R^2	0.795	0.968	0.800	0.968	0.795	0.968	0.970	0.971
Observations	273	271	273	271	273	271	240	240

Standard errors in parentheses clustered at year-month (min K=70). Coefficients are standardized to unit variances.

Average MPG and GHG Rating are multiplied by (-1) such that higher values respond to worse environmental performance.

+ p<0.10, * p<0.05, ** p<0.01, *** p<0.001

Transition risk Junior tranches are by design more exposed any types of risk. Junior tranches of auto ABS carry not just higher prepayment risk but also default risk, especially in the subprime market. Junior tranches are also exposed to potentially confounding transition risk. Transition risk would make higher-emission deals more risky. One would therefore expect an upward bias in β. Appendix Table A4 repeats the exercise in the junior tranches (B, C, and D tranche) of each deal. Odd columns in Appendix Table A4 show coefficients estimates without controlling for ex-post realizations of prepayment and default risk. The coefficients still indicate a significant brownium. Even columns add ex-post realizations of prepayment and default risk. In these specifications, the brownium flips sign and turns into an (insignificant) greenium, potentially driven by exposure to transition risk. Note, however, that the sample composition changes since many (prime) issuers tend to retain or privately place junior tranches. This result should thus be interpreted with care.

Specification Analysis Researcher often report only a subset of all reasonable specifications that test a given hypothesis. This can introduce a bias in which only specifications supporting the a priori hypothesis are reported. A Specification Curve Analysis (SCA, see Simonsohn et al., 2020) estimate all reasonable specifications and reports the results in a descriptive specification curve. Appendix Figure A3 and Figure A4 show SCAs for equation (3). The SCAs each test a set of 40 reasonable alternative specifications for both A-2 and A-3 tranches. The figures show the coefficients of β for the average financed tCO_2 emissions per vehicle and financed tCO_2 emissions per \$100,000, respectively. In only 3 out of 160 specifications is the β coefficient positive. In 103 out of 160 specifications is the brownium larger (but often noisier) than in column (4) of Panel (B) Table 3. Appendix Figure A3 and Figure A4 both show that the β coefficients are stable around the estimates of the main specification for wide range of plausible alternatives.

5.3 Discussion

The result that CO_2 emissions are not priced into issuance spreads indicates that ESG investors are not marginal in setting prices. That differences in greenness are not priced or even induce a brownium, stands in contrast to prior results in the literature. For example, Pástor et al. (2022) document a near-arbitrage opportunity in which German government green "twin" bonds relative to their non-green twins trade on average at 5bps lower yields. Similarly, Baker, Bergstresser, Serafeim and Wurgler (2022) and Zerbib (2019) estimate a greenium of 6bps and 2bps in green bonds, respectively.

There are several possible explanations for the difference between my results and prior results in the literature. I discuss the three most pertinent: (i) differences in issuance and market size,

(ii) high information acquisition cost, and (ii) "green window dressing".

Differences in issuance and market size could explain differences in pricing of green assets. Given that ESG capital is still scarcer than non-ESG capital, the chance of being the marginal buyer is decreasing in the issuance size. Several recent studies have looked at pricing differences in the green municipal-bond market (Larcker and Watts, 2020, Baker, Bergstresser, Serafeim and Wurgler, 2022). Painter (2020) reports that the average issuance size for municipal bonds over $1m is $13.4m. The average issuance size in the A-2 tranches of auto ABS I analyze is an order of magnitude larger at $367m. If ESG investors are infra-marginal in setting prices in auto ABS, we would still expect more greener assets on their balance sheet. I explore the balance sheets of ESG mutual funds in Section 6.

Second, high information acquisition could imply unawareness among ESG investor and thus explain the absence of a greenium. This, however, seems unlikely as there are free reports by rating agencies (e.g., the KBRA reports mentioned above) and ESG industry groups that discuss the greenness of auto ABS. Moreover, the emissions content of each ABS pool is directly related to the share of trucks it finances as I document in Table 1. This variable is featured in security prospectus and accessible in the loan-level data. The availability of free information and the presence of a sufficient statistic make it unlikely that information acquisition play a role in explaining the differences in findings.

Lastly, the ostensibly conflicting results can also be reconciled by a green window dressing effect in which securities that are especially marketed as green trade at a premium while securities that markedly differ in their true greenness but are not marketed as such do not. I leave the exploration of this channel for future research.

6 Mutual Funds Holdings

The result that CO_2 emissions are not priced into issuance spreads indicates that ESG investors are not marginal in setting prices. If ESG investors are infra-marginal in setting prices, their preferences for greener assets would still be reflected in greater quantities of greener assets. In seminal work, Hong and Kacperczyk (2009) argue that norm-constrained investors, like endowments or ESG mutual funds, are more likely to invest in responsible securities compared to unconstrained investors. I therefore turn to portfolio holdings of mutual funds to analyze if the CO_2 emissions of the collateral influence ESG mutual funds' decision to invest. The prospectuses of ESG mutual funds often detail their ESG approach for asset-backed securities:

> "[...] The fund's manager utilizes an investment process that combines fundamental

> credit research with analysis of ESG metrics. Environmental assessment involves issues such as carbon emissions, pollution, and renewable energy. [...] Potential asset-backed securities (ABS) are evaluated according to the manager's assessment of material ESG issues for the ABS sectors. The assessment utilizes sector specific metrics across ESG categories, insights from third-party data providers, our analysts' qualitative assessment and a sector-level risk evaluation.[...]"
>
> – ESG Mutual Fund Prospectus I

or

> "[...] When evaluating securitized debt securities (including RMBS, CMBS, and other asset-backed securities), the Adviser generally considers the issuer's ESG rating along with ESG factors related to the underlying pool of assets, such as energy efficiency and environmental impact of the underlying assets[...]"
>
> – ESG Mutual Fund Prospectus II

That fact that ESG mutual funds discuss "environmental impact of the underlying assets" indicates awareness of the differential impacts that collateral pools can have.

The rest of this section proceeds as follows. First, I discuss the N-PORT holdings data and describe the identification strategy. I then document two counterintuitive facts: ESG mutual funds (i) hold positions across the full distribution of CO_2 emissions and (ii) that ESG funds invest more in higher-emission deals. Lastly, I discuss these findings.

6.1 Data

I obtain mutual funds holdings from the SEC Form N-PORT over the period from 2019-Q3 to 2022-Q4. In the data I keep the first observation for which a mutual fund reports a position in an auto ABS. I trim portfolio shares at the 1% to account for outliers.[20] I observe 266 individual auto ABS deals in the holdings data.[21] Mutual funds are among the largest group of investors in auto ABS. The aggregate mutual funds complex holds on average 23% of total issuance size per deal on their balance sheets. For some senior tranches the share of issuance size that is held by mutual funds can go as high as 85%. Other notable institutional investors in senior auto ABS are insurance companies, corporate treasuries, and local governments. Table 5 provides summary statistics of the holdings data.

In the holdings data, I identify ESG mutual funds in two ways: (i) by their name using key

[20] The results are qualitatively similar to using 2%, 5%, or not trimming.

[21] The discrepancy to the 278 auto ABS deals in Section 5 is explained by a shorter sample period.

Table 5: Summary Statistics of Holdings Data

	Mean	SD	Median	Min	Max	N
Portfolio Share (%)	0.17	0.21	0.09	0.00	1.24	13,204
Coupon Yield	2.03	1.26	1.98	0.00	8.08	13,204
Tranche Size ($ m)	245.42	164.86	201.20	7.00	746.94	13,204
Weighted Average Life	2.54	1.03	2.56	0.11	5.28	13,204
Subprime ABS	0.47	0.50	0.00	0.00	1.00	13,204
Financed tCO2 per $100k	232.43	38.43	237.42	118.20	311.78	13,204
Avg. Fin. tCO2 per Vehicle	59.76	11.59	60.06	40.54	101.27	13,204

words[22] like 'sustainable', 'ESG', or 'climate' and (ii) using a list of "Sustainable Investment Mutual Funds and ETFs" offered by institutional member firms of "The Forum for Sustainable and Responsible Investing" (USSIF).[23] I identify 35 ESG funds (and 787 non-ESG funds) which hold at least one position in an auto ABS tranche over the sample period (32 ESG funds hold at least two positions). These ESG funds make up around 25% of the 143 ESG-bond funds which I am able to identify in all N-PORT filings.

6.2 Identification Strategy

To test if ESG funds tilt their portfolio towards greener auto ABS, I estimate a reduced form asset demand function similar in spirit to recent work by Koijen and Yogo (2019). I use the following reduced form model for portfolio shares in tranches t of auto ABS deal b by mutual funds i in year-quarter r:

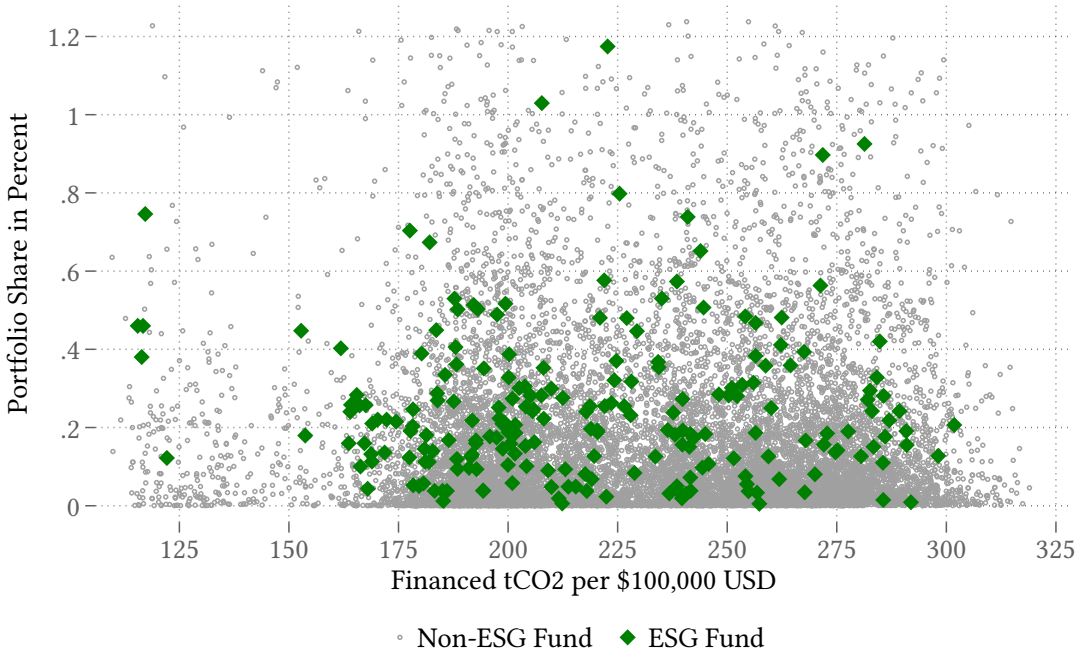
$$\text{Portfolio Share}_{itrb} = \alpha \left(\text{ESG Fund}_i \times \text{Green}_b\right) + \gamma_i + \gamma_b + \gamma_r + \zeta' X_t + \varepsilon_{itrb} \tag{4}$$

where Green_b is an indicator variable equal to one if the collateral pool's CO_2 emissions per vehicle are below the 25th percentile of all auto ABS deals; γ_i are fund fixed effects; γ_b are auto ABS deal fixed effects; and γ_r reporting year-quarter fixed effects. The vector X_t contains tranche characteristics which have different slopes for subprime and prime issues. These characteristics are the weighted average life, size, and annualized yield of the tranche.

The coefficient of interest is α which captures the effect of variation in greenness of the collateral on portfolio shares by ESG-funds relative to non-ESG funds. The stringent set of fixed

[22] Key word list: 'green', 'climate', 'esg', 'sustainable', 'enviroment', 'responsible', 'impact', 'catholic', 'social', 'sri', 'csr', 'community', and 'justice'.

[23] https://charts.ussif.org/mfpc/



Note: X-axis is jittered with normally distributed noise for readability.

Figure 3: Portfolio Shares of Mutual Funds in Auto ABS

effects implies that α is estimated from variation in greenness of auto ABS deals held by ESG funds relative to non-ESG funds during the same reporting period. Standard errors are clustered at the fund-level to account for common variation at the fund-level.

6.3 Results

Figure 3 plots portfolio shares in auto ABS of mutual funds against the financed CO_2 emissions per $100,000. The graph shows, non-parametrically, that ESG mutual funds hold positions across the full distribution of CO_2 emissions. That ESG funds have positions across the full distribution is surprising since common ESG strategies imply either outright exclusions of brown assets or best-in-class investment. Counterintuitive to best-in-class investment, Figure 3 also shows that ESG funds hold similar or higher shares in browner auto ABS.

Table 6 provides estimates of the relationship between greenness and ESG ownership using the parametric specification of equation (4). Panel A of Table 6 shows estimates for α using a green indicator variable. Column (1) of Panel A controls for tranche characteristics and absorbs fund and deal fixed effects. The α coefficient is identified from variation in greenness of auto ABS deals held by ESG funds relative to non-ESG funds. Column (2) adds reporting-quarter fixed

Table 6: Reduced Form Demand of Mutual Fund for Auto ABS

Panel A: Green Indicator

	(1) Portfolio Share	(2) Portfolio Share	(3) Portfolio Share	(4) Portfolio Share
Green (tCO2<p25)=1 × ESG Fund=1	-0.352**	-0.384**	-0.389**	-0.378***
	(0.122)	(0.122)	(0.121)	(0.109)
Tranche controls	Yes	Yes	Yes	Yes
Fund FE	Yes	Yes	Yes	Yes
Deal FE	Yes	Yes	Yes	No
Year-Quarter FE	No	Yes	Yes	Yes
Tranche FE	No	No	Yes	No
Cusip FE	No	No	No	Yes
Adj. R^2	0.653	0.657	0.659	0.707
Observations	13088	13088	13088	12977

Standard errors clustered at fund-level (min K=714). Coefficients are standardized to unit variances.

Panel B: other measures of relative *greenness*

	(1) Portfolio Share	(2) Portfolio Share	(3) Portfolio Share	(4) Portfolio Share
ESG Fund=1 × Avg. Fin. tCO2 per Vehicle	0.117*			
	(0.0508)			
ESG Fund=1 × Avg. MPG		0.185***		
		(0.0422)		
ESG Fund=1 × Truck Share			0.0724	
			(0.0680)	
ESG Fund=1 × Avg. GHG Rating (KBRA)				0.160***
				(0.0421)
Tranche controls	Yes	Yes	Yes	Yes
Fund FE	Yes	Yes	Yes	Yes
Deal FE	Yes	Yes	Yes	Yes
Year-Quarter FE	Yes	Yes	Yes	Yes
Tranche FE	Yes	Yes	Yes	Yes
Adj. R^2	0.659	0.659	0.659	0.652
Observations	13088	13088	13088	12285

Coefficients are standardized to unit variances. Average MPG and GHG Rating are negated.

Standard errors clustered at fund-level (min K=685).

+ p<0.10, * p<0.05, ** p<0.01, *** p<0.001

effects such that α restricted to variation within the same reporting period. Column (3) adds additional tranche FE effects to control for differential effects of across tranches (e.g., senior vs. junior). Finally, Column (4) uses Cusip fixed effects to control for unobserved heterogeneity at the security level.

The estimated coefficients of α are positive, indicating that the greenest auto ABS receive less capital from ESG funds. The estimates are stable across specifications and statistically significant (min $t = 2.89$). The estimates from Column (3) in Panel A of Table 6 indicate that ESG funds hold 0.08 percentage points less capital in the greenest 25% of auto ABS deals ($\approx 0.389 \times 0.21pp$). The magnitude of this difference is economically meaningful since the average portfolio share of ESG funds is 0.24%. ESG funds allocate 32% less capital to the greenest 25% of auto ABS deals.

Panel B of Table 6 presents similar estimates using other measures of relative greenness. All estimate of α in Panel B are identified from variation in greenness of auto ABS deals held by ESG funds relative to non-ESG funds during the same reporting period. The measure of greenness in Column (1) is the average financed tCO_2 per vehicle; average miles per gallon in Column (2); share of trucks in the collateral pool in Column (3); and the average GHG rating measure provided by KBRA in Column (4). In all columns the estimated coefficients of α are positive and of similar magnitude. For all but one measure they are statistically significant (min $t = 2.30$). The estimates in Column (1) of Panel B imply that going from the 10th percentile of average financed tCO_2 per vehicle to the 90th percentile (a 2.75 standard deviation distance equivalent to comparing Honda to Ford issuances) results in 28% higher portfolio shares of ESG compared to non-ESG funds ($\approx 0.117 \times 0.21 \times 2.75 \times 0.24^{-1}$).

6.4 Discussion

This subsection discusses the results that ESG mutual funds (i) hold positions across the full distribution of CO_2 emissions and (ii) that ESG funds invest more in higher-emission deals. Also discussed are the implications for retail investors of ESG mutual funds, fund managers, and policy makers in the finance and ESG investing communities.

The two findings, that ESG funds invest across the full distribution of CO_2 emissions and seem to invest more in higher-emissions deals, are hard to reconcile with standard ESG strategies.

There is an active debate about which strategy ESG investors should follow: *exit* (divestment or exclusion) versus *voice* (shareholder activism). Investments in debt securities are restricted to exit since creditors have no voting rights.[24] The most popular exit strategies are outright

[24]This is especially the case for auto ABS since they are issued by special purpose vehicles which are bankruptcy-remote from their sponsoring entity.

exclusion or best-in-class investment. Edmans et al. (2022) analyze whether exclusion or best-in-class investment is more effective and find that best-in-class is more effective since it provides incentives to take corrective actions. Both strategies are inconsistent with the observed pattern that ESG funds are invested across the whole distribution of CO_2 emissions intensity and that they allocate more capital towards higher emissions deals compared to non-ESG funds.

High information acquisition cost could deter ESG mutual funds from performing the necessary analysis to be able to distinguish brown from green auto ABS. This, however, seems unlikely as there are free reports by rating agencies (e.g., the KBRA reports mentioned above) and ESG industry groups that discuss the greenness of auto ABS. Moreover, the emissions content of each ABS pool is directly related to the share of trucks it finances as I document in Table 1. This variable is featured in security prospectus and accessible in the loan-level data. The availability of free information and the presence of a "sufficient statistic" make it unlikely that information acquisition cost play a significant role in explaining the holdings pattern of ESG mutual funds.

The holdings results have implications for retail investors of ESG mutual funds in ESG mutual funds, as they suggest that ESG mutual funds may not be fulfilling their stated objectives of promoting environmentally sustainable investing. Since investors in ESG mutual funds pay higher fees to have portfolios aligned with ESG considerations (Baker, Egan and Sarkar, 2022) and given that mutual funds have a fiduciary duty to promote the best interest of their clients, ESG fund investors can expect their money to be allocated with sensitivity toward ESG matters.

The implications for fund managers in the light of these findings could be significant. The findings suggest that ESG mutual funds are not investing in the most environmentally-friendly securities, which could raise questions about the effectiveness of their investment strategies and veracity of their stated environmental approaches. Fund managers may need to re-evaluate their investment processes and methods to ensure that they are fulfilling their obligations to promote environmentally sustainable investing. Fund managers may need to look for alternative investments that are better aligned with their ESG objectives and consider the environmental impact of their investments more closely in order to meet the expectations of their investors.

The findings that ESG funds hold positions across the full distribution of CO_2 emissions, and tend to hold higher portfolio shares in high-emissions issues, suggest that there may be a need for more clarity and transparency in ESG labeling and investment processes. Policymakers may need to provide greater guidance to the financial sector on what constitutes environmentally sustainable investing, and to ensure that ESG labels accurately reflect the environmental impact of investments.

ESG regulation in the United States is still in its infancy. The SEC has taken some steps to

provide guidance to the financial industry to ensure that ESG labels accurately reflect the environmental impact of investments. The SEC has issued several statements and guidance documents related to ESG investing and has encouraged companies to provide more comprehensive and transparent disclosure of their ESG practices and impact. In Europe, there are similar efforts to ensure the accuracy of ESG labeling. The European Union has adopted the Sustainable Finance Disclosure Regulation (SFDR), which requires companies and financial market participants to provide more comprehensive and transparent disclosure of their sustainability risks, impacts, and objectives. The SFDR also establishes a harmonized framework for ESG labeling and ensures that investors have access to accurate and relevant information about the environmental and social impact of their investments.

7 Conclusion

This paper sheds new light on the question of whether a greenium exists in debt markets. The study compares the cost of capital for green and brown automobile asset-backed securities and provides a causal estimate of the premium investors are willing to pay for greener securities. The unique features of securitized auto loans circumvent the identification issues commonly raised in ESG investing studies. The results show a *brownium*, a negative greenium, of 5 to 10 basis points for senior tranches in the auto ABS market. The brownium is robust to choices of alternative measures of greenness, samples, and specifications.

The paper further provides evidence that ESG mutual funds hold positions across the full distribution of CO_2 emissions, indicating that they are not investing in the most environmentally-friendly securities. The paper further shows that ESG funds tend to hold higher portfolio shares in high-emissions issues compared to non-ESG funds. This is surprising since greener auto ABS are not riskier and neither do browner securities pay higher yields. Both findings are hard to reconcile with strategies associated with ESG investing.

The findings have implications for retail investors of ESG mutual funds, fund managers, and policy makers. The results suggest that ESG mutual funds are not investing in the most environmentally-friendly securities, which raises questions about the effectiveness of ESG investment strategies.

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A Appendix

(a) Santander SDRIVE 2021-4 Subprime Issue

Pricing $1.8bn Santander Drive Auto Receivables Trust 2021-4

Issuer: Santander Consumer USA
Lead Managers: Citi(str.), JPM, and SIS
DE&I Co-managers: AmeriVet Securities, Great Pacific Securities, Mischler Financial Group

Anticipated Capital Structure:

CL	OFF. AMT	WAL	F/M	L.FNL	BENCH	SPRD	YLD%	CPN	PX
A-1	$222.40	0.13	F1+/P-1	11/15/2022	Intl.	7	0.16802	0.16802	100.00000
A-2	$543.10	0.64	AAA/Aaa	08/15/2024	EDSF	20	0.380	0.37	99.99378
A-3	$292.37	1.43	AAA/Aaa	08/15/2025	EDSF	15	0.517	0.51	99.99081
B	$288.61	2.10	AA/Aaa	06/15/2026	IntS	30	0.887	0.88	99.98887
C	$243.48	2.80	A/Aa2	02/16/2027	IntS	47	1.271	1.26	99.97902
D	$241.38	3.58	BBB/Baa2	10/15/2027	IntS	70	1.685	1.67	99.98569
E	$131.18	3.97	NR/B2	<<NOT OFFERED>>					

(b) CarMax 2019-1 Prime Issue

$1.5bln CarMax (CARMX) 2019-1
JOINT BOOKRUNNERS : Credit Suisse (str), Barclays, Wells Fargo
CO-MANAGERS : MUFG, Scotia, SMBC, TD

CLS	$AMT(MM)	WAL	S&P/FITCH	P.WIN	L.FNL	BNCH	SPRD	YLD%
A1	277.000	0.28	A-1+/F1+	1-7	01/2020	IntL -	1	2.78007
A2A	412.000	1.16	AAA/AAA	7-22	07/2022	EDSF +	31	3.045
A2B	100.000	1.16	AAA/AAA	7-22	07/2022	1mL +	31	
A3	493.900	2.64	AAA/AAA	22-43	03/2024	IntS +	40	3.074
A4	107.910	3.84	AAA/AAA	43-48	08/2024	IntS +	65	3.283
B	42.170	3.98	AA/AA	48-48	11/2024	IntS +	85	3.479
C	39.910	3.98	A/A	48-48	01/2025	IntS +	115	3.779
D	27.110	3.98	BBB/BBB	48-48	08/2025	IntS +	145	4.079

* Exp. Settle: 01/23/19
* First Pay Date: 02/15/19
* Px Speed: 1.30% ABS to 10% Call
* Timing: PRICED
* Format: Public/SEC
* ERISA: Yes
* Min Denoms: $5k by $1k
* B&D: Credit Suisse

Figure A1: Examples of Typical auto ABS Deals

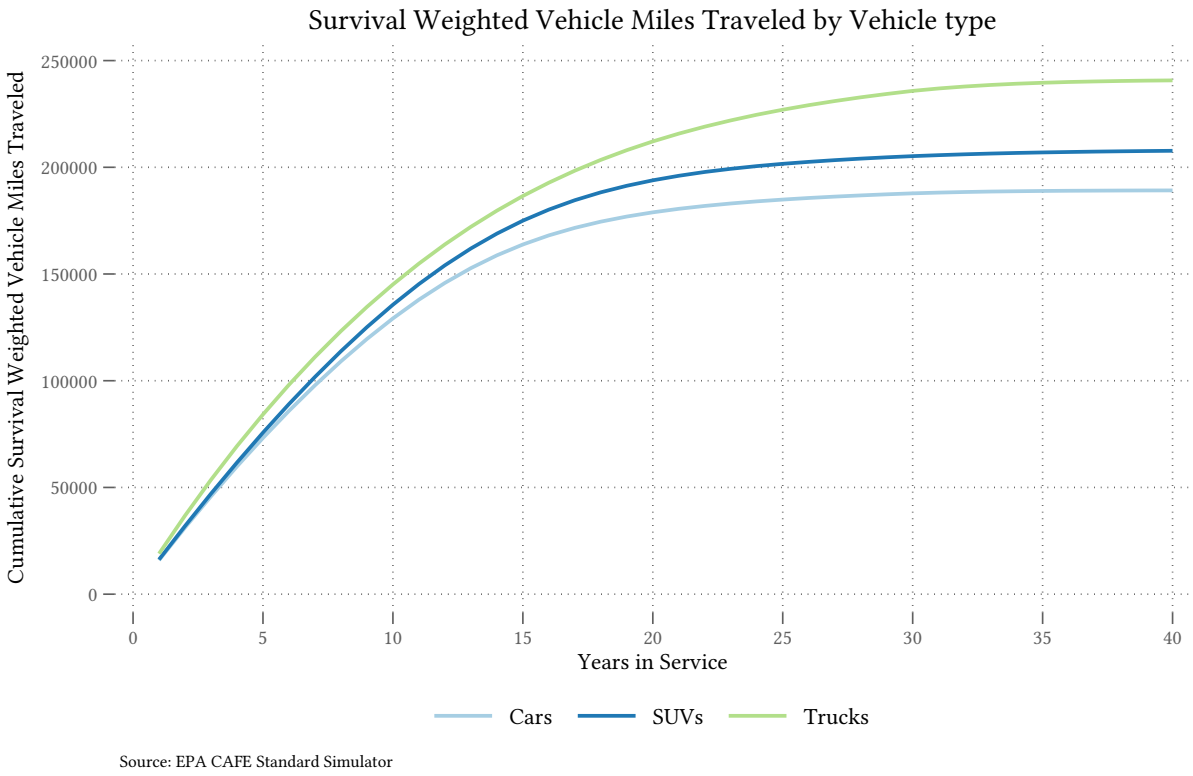


Figure A2: Survival-Weighted Vehicle Miles by Vehicle Type

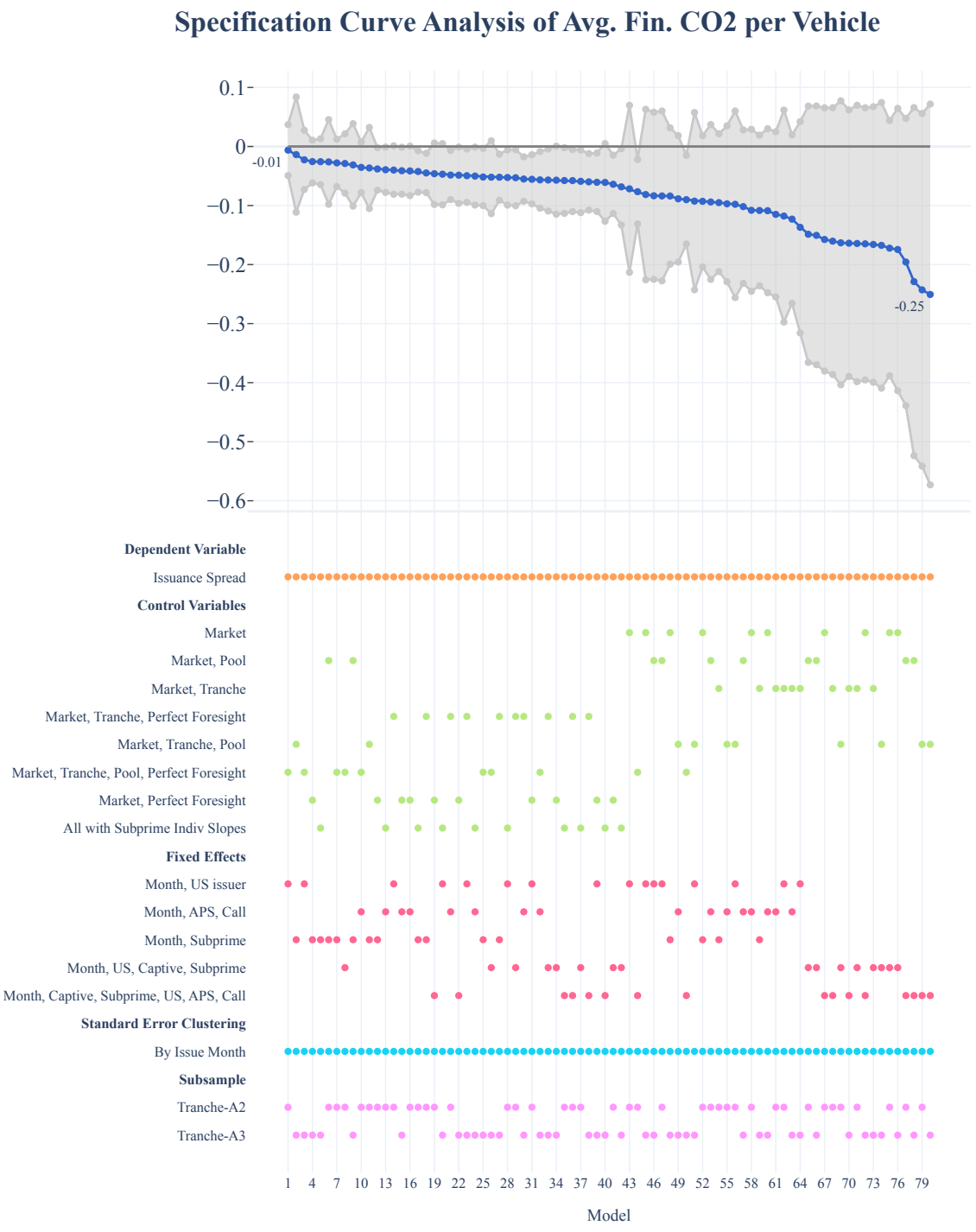


Figure A3: Specification Curve Analysis: Avg. Fin. tCO_2 per Vehicle

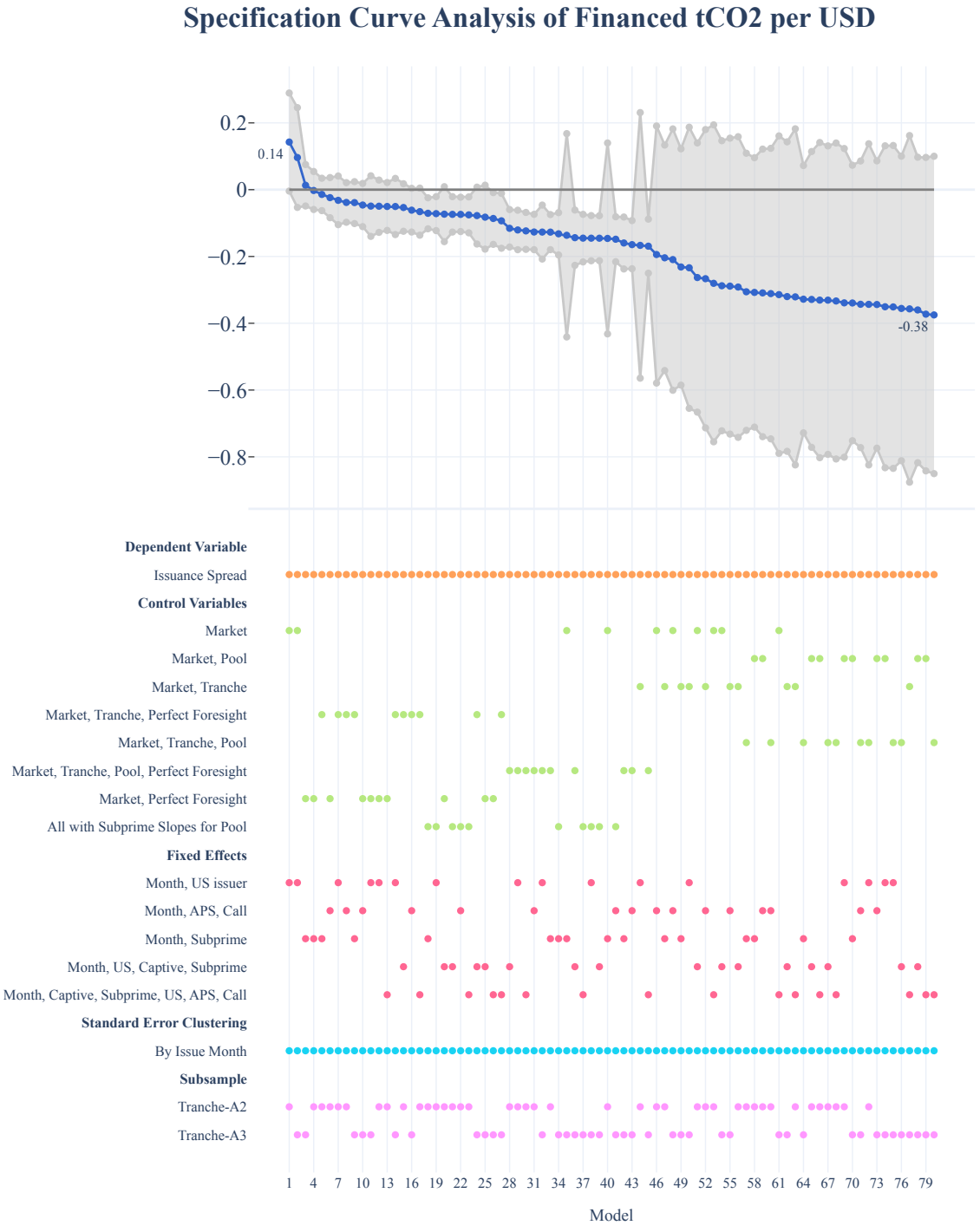


Figure A4: Specification Curve Analysis: tCO_2 per \$100,000.

Table A1: Summary Statistics Loan-Level Data

	Mean	SD	Median	Min	Max	N
Original Interest Rate	7.62	6.96	4.99	0.00	30.00	18,768,831
Original Loan Amount ($)	26,142.09	12,500.23	23,900.04	518.03	248,681.95	18,768,832
Original Loan Term (months)	67.53	8.68	72.00	7.00	96.00	18,768,832
Credit Score	711.79	101.26	723.00	250.00	900.00	17,935,867
Payment-to-Income Ratio	0.08	0.05	0.08	0.00	0.79	18,550,004
Income Verified	0.09	0.29	0.00	0.00	1.00	18,768,832
Loan-to-Value	0.90	0.16	1.00	0.01	1.00	18,767,550
Outstanding Balance Ratio	0.83	0.24	0.93	0.00	1.00	18,768,828
Vehicle Value Amount ($)	27,732.79	13,378.74	25,250.00	0.00	1,084,455.00	18,768,829
Vehicle Age (Years)	2.70	2.53	2.00	0.00	35.00	18,768,832
Used Vehicle	0.46	0.50	0.00	0.00	1.00	18,768,832
SVM, Financed	162,450.37	39,872.01	173,251.82	254.15	240,728.61	18,768,832
SVM, Total	202,962.96	17,249.52	207,738.97	189,173.82	240,728.61	18,768,832
tCO2, total Lifetime	78.93	33.12	72.55	0.00	538.75	18,768,832
tCO2, remaining Lifetime	63.00	31.95	56.71	0.00	538.75	18,768,832
tCO2, financed remaining Lifetime	47.03	29.56	44.58	0.00	538.75	18,767,546

Table A2: Covariate Balance Test of *Green* and *Brown* auto ABS Deals

Variable	(1) Brown (CO2>p25) Mean/(SE)	(2) Green (CO2<=p25) Mean/(SE)	(1)-(2) Pairwise t-test Mean difference
Financed tCO2 per $100,000	235.567	173.951	61.616***
	(2.238)	(2.430)	
Yield Curve 12m	0.015	0.015	0.000
	(0.001)	(0.002)	
VIX	18.222	23.185	-4.963
	(0.486)	(0.844)	
5 Year Breakeven Inflation	1.944	2.293	-0.349***
	(0.036)	(0.087)	
Weight. Avg. Life	0.943	1.097	-0.153
	(0.019)	(0.052)	
Attachment Point	0.493	0.458	0.035
	(0.005)	(0.006)	
Loan-to-Value	0.922	0.904	0.018
	(0.003)	(0.004)	
Mean Credit Score	691.689	753.165	-61.476***
	(5.358)	(5.320)	
Income Verified	0.122	0.114	0.008
	(0.017)	(0.037)	
Mean Interest Rate	8.577	4.613	3.964
	(0.424)	(0.454)	
Used Vehicles Share	0.443	0.332	0.112
	(0.023)	(0.042)	
Mean % of outstanding	0.909	0.867	0.042
	(0.005)	(0.008)	
Remaining Term (Months)	60.933	56.583	4.351*
	(0.469)	(0.830)	
Delta Realized ABS	-0.018	-0.044	0.026
	(0.018)	(0.025)	
Realized % Loans Delinquent 30+	2.967	0.743	2.224
	(0.217)	(0.161)	
Number of observations	207	68	271

Note: Similar to the regression analysis the t-test includes clean-up call FE, APS FE, captive and subprime FE, and year-month FE.

Table A3: Estimates of the Greenium using other Senior Tranches

	(1) Issuance Spread	(2) Issuance Spread	(3) Issuance Spread	(4) Issuance Spread	(5) Issuance Spread	(6) Issuance Spread
	A-1 Tranche		A-3 Tranche		A-4 Tranche	
Financed tCO2 per $100k	-0.159	-0.0102	-0.229*	-0.145***	-0.0929**	-0.144***
	(0.112)	(0.0380)	(0.113)	(0.0357)	(0.0314)	(0.0379)
Delta Realized ABS		-0.0465+		-0.140***		-0.0757*
		(0.0271)		(0.0291)		(0.0363)
Realized % Loans Delinquent 30-60 days		0.109		-0.117		-0.0832
		(0.108)		(0.133)		(0.0687)
Year-Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Market condition controls	Yes	Yes	Yes	Yes	Yes	Yes
Subprime, captive FEs	Yes	Yes	Yes	Yes	Yes	Yes
Pre-payment speed controls	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.756	0.954	0.794	0.960	0.963	0.962
Observations	234	233	269	267	190	189

Standard errors in parentheses clustered at year-month. Coefficients are standardized to unit variances.
+ p<0.10, * p<0.05, ** p<0.01, *** p<0.001

Table A4: Estimates of the Greenium using Junior Tranches

	(1) Issuance Spread	(2) Issuance Spread	(3) Issuance Spread	(4) Issuance Spread	(5) Issuance Spread	(6) Issuance Spread
	B Tranche		C Tranche		D Tranche	
Financed tCO2 per $100k	-0.577+	0.0233	-0.516+	-0.117	-1.124***	0.138
	(0.320)	(0.0779)	(0.270)	(0.0816)	(0.168)	(0.168)
Delta Realized ABS		0.00342		-0.0358		-0.223
		(0.0514)		(0.0513)		(0.135)
Realized % Loans Delinquent 30-60 days		-0.308		-0.484*		-0.0190
		(0.239)		(0.189)		(0.233)
Year-Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Market condition controls	Yes	Yes	Yes	Yes	Yes	Yes
Subprime, captive FEs	Yes	Yes	Yes	Yes	Yes	Yes
Pre-payment speed controls	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.862	0.968	0.928	0.980	0.995	0.999
Observations	171	170	153	152	72	71

Standard errors in parentheses clustered at year-month. Coefficients are standardized to unit variances.
+ p<0.10, * p<0.05, ** p<0.01, *** p<0.001